

GearDesign Report

Cylindrical gear(pair)

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1. Summary

1.1. Gear geometry

Meshing pair: Gear1 - Gear2

	Symbol	Pair1 Gear1	Pair1 Gear2
Normal module (mm)	m_n	2.0000	
Pressure angle at normal section (deg)	α_n	20.0000	
Helix angle at reference circle (deg)	β	25.0000	
Manual normal backlash (mm)	j_{bn}	0.0000	
Number of teeth	z	35.0000	76.0000
Facewidth (mm)	b	35.0000	35.0000
Profile shift coefficient	x	0.0000	-0.0000
Quality (ISO)	Q	6.0000	6.0000
Center distance (mm)	a	122.4750	
Tooth thickness tolerance	-	DIN 3967 cd25	DIN 3967 cd25
Gear material	-	DIN 18CrNiMo7 ($\sigma_{Hlim} = 430$ Mpa)	DIN 18CrNiMo7 ($\sigma_{Hlim} = 430$ Mpa)
Reference diameter (mm)	d	77.2360	167.7130
Tip diameter (mm)	d_a	81.236 (81.236 / 81.136)	171.713 (171.713 / 171.613)
Root diameter (mm)	d_f	72.236 (72.044 / 71.934)	162.713 (162.452 / 162.315)
Number of teeth spanned	k	6.0000	12.0000
Base tangent length (no backlash) (mm)	W_k	33.770 (33.704 / 33.667)	70.715 (70.626 / 70.579)
Effective Diameter of ball/pin (mm)	D_M	3.5000	3.5000
Diametral measurement over two balls (mm)	M_{dk}	82.172 (81.998 / 81.898)	172.772 (172.523 / 172.392)

1.2. Total safety factor

	Gear1 [Left]	Gear1 [Right]	Gear2 [Left]	Gear2 [Right]	Required S.F
Contact S.F	1.3688		1.4017		1.0000
Bending S.F	2.3182		2.2968		1.4000
Scuffing S.F (Flash Temp.)	4.5952				2.0000
Micropitting S.F	2.2154				1.0000

1.3. Total damage & life

	Gear1 [Left]	Gear1 [Right]	Gear2 [Left]	Gear2 [Right]
Contact Damage [%]	0.0000	0.0000	0.0000	0.0000
Bending Damage [%]	0.0000	0.0000	0.0000	0.0000
Contact Life [hr]	∞	∞	∞	∞
Bending Life [hr]	∞	∞	∞	∞
Total Req. Life [hr]	10000.0000			

1.4. Others

	Gear1 - Gear2
Transverse Contact Ratio	1.4463
Overlap Ratio	2.3542
Total Contact Ratio	3.8005
Worst Pitch Line Velocity [m/s]	40.4409
Total Efficiency [%]	97.7146
Worst Mesh Loss [W]	407.2432
Worst Churning Loss [W]	3020.8387
Worst Peak-to-Peak T.E. [μm]	0.5160
Gear Total Mass [kg]	1.7219
Hunting tooth	Complete

1.5. Summary - Load cases

[illegible]

2. Detail geometry

2.1. Input gear geometry

Meshing pair: Gear1 - Gear2

	Symbol	Pair1 Gear1	Pair1 Gear2
Normal module (mm)	m_n	2.0000	
Pressure angle at normal section (deg)	α_n	20.0000	
Helix angle at reference circle (deg)	β	25.0000	
Manual normal backlash (mm)	j_{bn}	0.0000	
Number of teeth	z	35.0000	76.0000
Facewidth (mm)	b	35.0000	35.0000
Profile shift coefficient	x	0.0000	-0.0000
Quality (ISO)	Q	6.0000	6.0000
Center distance (mm)	a	122.4749	
Tooth thickness tolerance	-	DIN 3967 cd25	DIN 3967 cd25
Addendum coefficient	h^*_aP	1.0000	1.0000
Dedendum coefficient	h^*_fP	1.2500	1.2500
Root radius coefficient	ρ^*_fP	0.3800	0.3800
Tip form height coefficient	h^*_{FaP}	0.9500	0.9500
Ramp angle (deg)	α_{KP}	50.0000	50.0000
Protuberance height coefficient	h^*_{prP}	0.0000	0.0000
Protuberance angle (deg)	α_{prP}	0.0000	0.0000
Tip alteration (mm)	k_{m_n}	0.0000	0.0000
Tip radius coefficient (for topping)	ρ^*_{aP}	0.0000	0.0000
Gear material	-	DIN 18CrNiMo7 ($\sigma_{Hlim} = 430$ Mpa)	DIN 18CrNiMo7 ($\sigma_{Hlim} = 430$ Mpa)
Young's modulus (N/mm ²)	E	2.0600×10^5	2.0600×10^5
Poisson's ratio	ν	0.3000	0.3000
Density (kg/m ³)	ρ	7830.0000	7830.0000
Coeff. of thermal expansion (10 ⁻⁸ /°C)	α_{coeff}	11.5000	11.5000
Surface hardness	HRC	61.0000	61.0000
Core hardness	HBW	325.0000	325.0000
Tensile strength (N/mm ²)	σ_B	1200.0000	1200.0000
Yield strength (N/mm ²)	σ_s	850.0000	850.0000
Specific heat capacity (J/kg.K)	c_M	485.0000	485.0000
Specific heat conductivity (W/m.K)	λ_M	50.0000	50.0000
Mean roughness height, root (um)	R_{ZF}	20.0000	20.0000
Mean roughness height, flank (um)	R_{ZH}	4.8000	4.8000
Roughness average value, root (um)	R_{AF}	3.0000	3.0000
Roughness average value, flank (um)	R_{AH}	0.6000	0.6000

	Symbol	Pair1 Gear1	Pair1 Gear2
Fatigue strength for bending stress (N/mm ²)	σ_{Flim}	430.0000	430.0000
Fatigue strength for contact stress (N/mm ²)	σ_{Hlim}	1500.0000	1500.0000

2.2. Calculated geometry - Gear

Meshing pair: Gear1 - Gear2

	Symbol	Pair1 Gear1	Pair1 Gear2
Gear ratio	GR	2.1714	
Transverse module (mm)	m_t	2.2068	
Working normal pressure angle (deg)	α_{wn}	20.0000	
Transverse pressure angle (deg)	α_t	21.8802	
Working transverse pressure angle (deg)	α_{wt}	21.8802	
Working transverse pressure angle, e/i (deg)	$\alpha_{wt}, e/i$	21.8948 / 21.8657	
Base helix angle (deg)	β_b	23.3990	
Helix angle at operating pitch circle (deg)	β_w	25.0000	
Radial backlash without tolerance (mm)	j_{r0}	0.0000	
Cumferential backlash without tolerance (mm)	j_{t0}	0.0000	
Min/max circumferential backlash by flank tolerance for each gear (mm)	$j_t, e/i$	0.0772 / 0.1214	0.1048 / 0.1600
Min/max normal backlash by flank tolerance for each gear (mm)	$j_n, e/i$	0.0658 / 0.1034	0.0893 / 0.1363
Cumferential backlash, e/i (mm)	$j_{tw}, e/i$	0.1720 / 0.2914	
Normal backlash, e/i (mm)	$j_{nw}, e/i$	0.1465 / 0.2482	
Angluar backlash, min (deg)	j_θ, min	0.2552	0.1175
Angluar backlash, max (deg)	j_θ, max	0.4323	0.1991
Effective facewidth (mm)	b_{eff}	35.0000	
Sum of the profile shift coefficients	Sum_x	-0.0000	
Generating profile shift coefficient	$x_E, e/i$	-0.0481 / -0.0756	-0.0653 / -0.0996
Prevent undercut profile shift coefficient	x_{Emin}	-1.6817	-4.8231
Max. profile shift coeff. (min top land)	x_{Emax}	1.7963	2.9107
Reference center distance, $x_1=x_2=0$ (mm)	a_d	122.4749	
Working center distance (mm)	a_w	122.4749	
Reference diameter (mm)	d	77.2365	167.7134
Operating pitch diameter (mm)	d_w	77.2365	167.7134
Operating pitch diameter with tolerance (mm)	$d_w, e/i$	77.2443 / 77.2286	167.7306 / 167.6963
27		81.2365	171.7134
Tip diameter with tolerance (mm)	$d_a, e/i$	81.2365 / 81.1365	171.7134 / 171.6134

	Symbol	Pair1 Gear1	Pair1 Gear2
Root diamter (mm)	d_f	72.2365	162.7134
Root diameter with tolerance (mm)	d_f, e/i	72.0441 / 71.9342	162.4524 / 162.3151
Tip form diameter (mm)	d_Fa	81.0365	171.5134
Tip form diameter with tolerance (mm)	d_Fa, e/i	80.8293 / 80.7112	171.2425 / 171.1001
Root form diamter (mm)	d_Ff	73.9107	164.0163
Root form diameter with tolerance (mm)	d_Ff, e/i	73.7864 / 73.7169	163.7965 / 163.6820
Active tip diameter (EAP) (mm)	d_Na	81.0365	171.5134
Active tip diameter with tolerance (mm)	d_Na, e/i	80.8293 / 80.7112	171.2425 / 171.1001
Active root diameter (SAP) (mm)	d_Nf	74.2012	164.5615
Active root diameter with tolerance (mm)	d_Nf, e/i	74.4816 / 74.3535	164.8130 / 164.6850
Mean diameter for mass calculation (mm)	d_m	76.5878	167.0236
Gear mass (kg)	m	0.5064	1.2155
Gear inertia (kg.m^2)	J	0.0000	0.0000
Nominal tip clearance (mm)	c	0.5000	0.5000
Effective tip clearance, e/i (mm)	c, e/i	0.7617 / 0.6180	0.7136 / 0.5837
Min. margin of involute contact, (d_Nf_i - d_Ff_e) / 2 (mm)	c_Nff	0.2835	0.4442

2.3. Calculated geometry - Tooth

Meshing pair: Gear1 - Gear2

	Symbol	Pair1 Gear1	Pair1 Gear2
Tooth depth (mm)	h	4.5000	4.5000
Tooth depth, e/i (mm)	h, e/i	4.6511 / 4.5462	4.6992 / 4.5805
Working depth (mm)	h_w	4.0000	
Addendum (mm)	h_a	2.0000	2.0000
Addendum, e/i (mm)	h_a, e/i	2.0000 / 1.9500	2.0000 / 1.9500
Dedendum (mm)	h_f	2.5000	2.5000
Dedendum, e/i (mm)	h_f, e/i	2.5962 / 2.6511	2.6305 / 2.6992
Number of virtual gear teeth	z_n	45.8493	99.5584
Tooth thickness coeff.	s_nc	1.5708	1.5708
Transverse tooth thickness (mm)	s_t	3.4664	3.4664
Normal tooth thickness (mm)	s_n	3.1416	3.1416
Normal tooth thickness at d_a (mm)	s_an	1.5441	1.6160
Normal tooth thickness at d_a, e/i (mm)	s_an, e/i	1.5174 / 1.4295	1.5604 / 1.4682
Normal tooth thickness at d_Fa (mm)	s_Fan	1.6362	1.6983
Normal tooth thickness at d_Fa, e/i (mm)	s_Fan, e/i	1.7109 / 1.6162	1.7699 / 1.6613
Normal space width at d_f (mm)	e_fn	1.7097	1.4710
Normal space width at d_f, e/i (mm)	e_fn, e/i	1.7518 / 1.7778	1.4897 / 1.5001

	Symbol	Pair1 Gear1	Pair1 Gear2
Normal pitch (mm)	p_n	6.2832	
Transverse pitch (mm)	p_t	6.9327	
Normal base pitch (mm)	p_bn	5.9043	
Transverse base pitch (mm)	p_bt	6.4333	
Axial pitch (mm)	p_x	14.8673	
Lead (mm)	p_z	520.3549	1129.9135

2.4. Calculated geometry - Mesh

Meshing pair: Gear1 - Gear2

	Symbol	Pair1 Gear1	Pair1 Gear2
Length of the path of contact (A to E) (mm)	g_α	9.3046	
Length of path of contact, e/i (mm)	g_α, e/i	8.7915 / 8.4255	
Length of approach POC of pinion (A to C) (mm)	g_f1	4.7892	
Length of approach POC of pinion, e/i (mm)	g_f1, e/i	4.4885 / 4.4426	
Length of recess POC of pinion (C to E) (mm)	g_a1	4.5154	
Length of recess POC of pinion, e/i (mm)	g_a1, e/i	4.3030 / 4.2819	
Length T1 - A (mm)	T1A	9.6026	
T1A tolerance (mm)	e/m/i	10.1307 / 10.0117 / 9.8926	
Length T1 - B (mm)	T1B	12.4739	
T1B tolerance (mm)	e/m/i	12.2509 / 12.1868 / 12.1228	
Length T1 - C (mm)	T1C	14.3918	
T1C tolerance (mm)	e/m/i	14.4023 / 14.3918 / 14.3812	
Length T1 - D (mm)	T1D	16.0359	
T1D tolerance (mm)	e/m/i	16.5640 / 16.4450 / 16.3260	
Length T1 - E (mm)	T1E	18.9072	
T1E tolerance (mm)	e/m/i	18.6842 / 18.6202 / 18.5562	
Length T2 - A (mm)	T2A	36.0399	
T2A tolerance (mm)	e/m/i	35.7163 / 35.6308 / 35.5453	
Length T2 - B (mm)	T2B	33.1686	
T2B tolerance (mm)	e/m/i	33.5531 / 33.4556 / 33.3580	
Length T2 - C (mm)	T2C	31.2507	
T2C tolerance (mm)	e/m/i	31.2736 / 31.2507 / 31.2277	
Length T2 - D (mm)	T2D	29.6065	
T2D tolerance (mm)	e/m/i	29.2829 / 29.1974 / 29.1120	
Length T2 - E (mm)	T2E	26.7353	
T2E tolerance (mm)	e/m/i	27.1198 / 27.0223 / 26.9247	
Length T1 - T2 (mm)	T1T2	45.6425	

	Symbol	Pair1 Gear1	Pair1 Gear2
T1T2 tolerance (mm)	e/m/i	45.6760 / 45.6424 / 45.6089	
Diameter of single contact point A (mm)	d_A	74.2012	171.5134
Diameter of single contact point B (mm)	d_B	75.8905	169.1804
Diameter of single contact point C (mm)	d_C	77.2365	167.7134
Diameter of single contact point D (mm)	d_D	78.5212	166.5160
Diameter of single contact point E (mm)	d_E	81.0365	164.5615
Transverse contact ratio	ϵ_{α}	1.4463	
Overlap ratio	ϵ_{β}	2.3542	
Total contact ratio	ϵ_{γ}	3.8005	
Total contact ratio with tolerance	e/m/i	3.721 / 3.692 / 3.664	
Specific sliding at Tip	ζ_a	0.3488	0.4214
Specific sliding at Root	ζ_f	-0.7284	-0.5356

2.5. Calculated geometry - Accuracy

Meshing pair: Gear1 - Gear2

	Symbol	Pair1 Gear1	Pair1 Gear2
Quality Standard	-	ISO 1328: 2013	ISO 1328: 2013
Single pitch deviation (μm)	f_pt	8.5000	8.5000
Base circle pitch deviation (μm)	f_pb	7.9000	7.9000
Sector pitch deviation over k/8 pitches (μm)	F_pk/8T	17.0000	20.0000
Profile form deviation (μm)	f_fa	8.5000	8.5000
Profile slope deviation (μm)	f_Ha	7.0000	7.0000
Total profile deviation (μm)	F_α	11.0000	11.0000
Helix form deviation (μm)	f_fb	10.0000	11.0000
Helix slope deviation (μm)	f_Hβ	9.0000	9.5000
Total helix deviation (μm)	F_β	13.0000	15.0000
Cumulative pith deviation, total (μm)	F_p	26.0000	29.0000
Runout (μm)	F_r	23.0000	26.0000
Single flank composite, tooth-to-tooth (μm)	f_isT	8.0000	8.0000
Single flank composite, total (μm)	F_isT	34.0000	37.0000
Tooth-to-tooth radial composite deviation (μm)	f_idT	9.5000	9.5000
Total radial composite deviation (μm)	F_idT	31.0000	37.0000

2.6. Calculated geometry - Measurement

Meshing pair: Gear1 - Gear2

	Symbol	Pair1 Gear1	Pair1 Gear2
Number of teeth spanned	k	6.0000	12.0000
Base tangent length (mm)	W_k	33.7702	70.7149
Base tangent length with tolerance (mm)	W_k, e/i	33.7045 / 33.6669	70.6257 / 70.5787
Tolerance of W_k (mm)	ΔW_k , e/i	-0.0658 / -0.1034	-0.0893 / -0.1363
Diameter of contact point (mm)	d_M	78.0560	168.5820
Theoretical diameter of ball/pin (mm)	D_M.theory	3.3844	3.3617
Effective Diameter of ball/pin (mm)	D_M.eff	3.5000	3.5000
Radial single ball measurement (mm)	M_rk	41.1256	86.3858
Radial single ball measurement with tolerance (mm)	M_rk, e/i	41.0387 / 40.9886	86.2617 / 86.1959
Diametral two ball measurement (mm)	M_dk	82.1720	172.7716
Diametral two ball measurement with tolerance (mm)	M_dk, e/i	81.9982 / 81.8981	172.5233 / 172.3918
Tolerance of M_dk (mm)	ΔM_{dk}	-0.1737 / -0.2739	-0.2483 / -0.3799

3. Detail rating

3.1. General

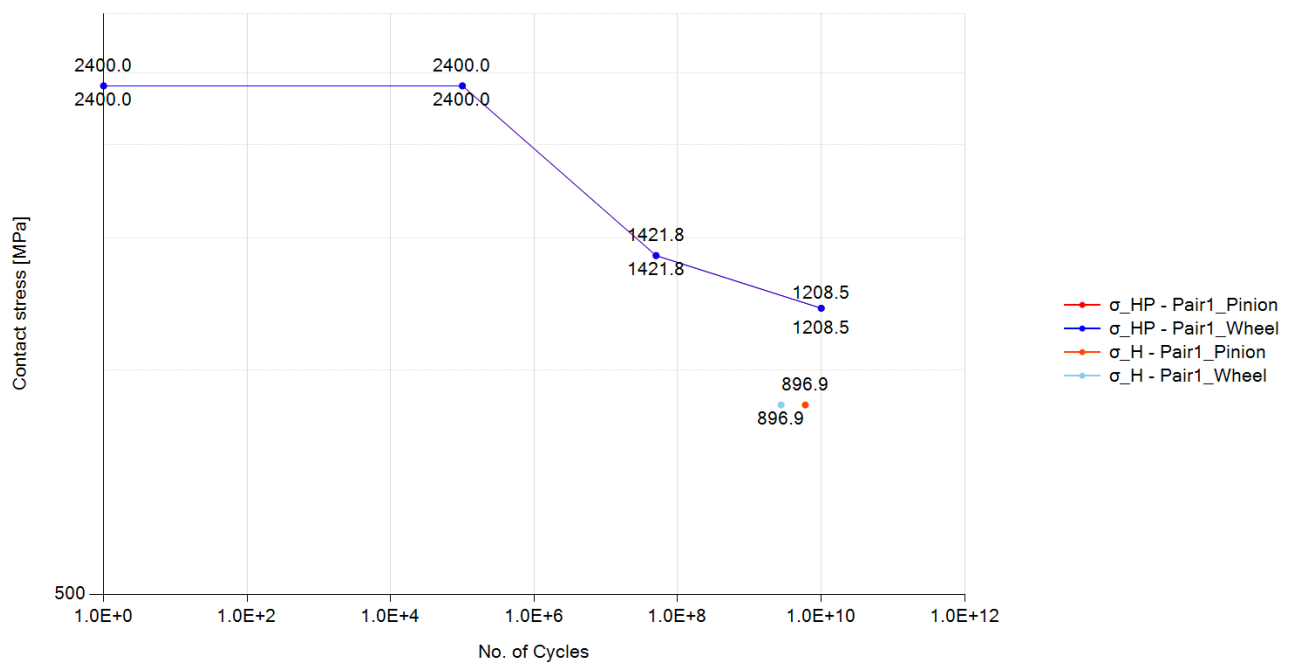
Load case 1, Meshing pair: Gear1 - Gear2

	Symbol	Pair1 Gear1	Pair1 Gear2
Rotational speed (rpm)	ω	10000.0000	4605.2630
Nominal tangential load at d (N)	F_t	3709.1150	
Nominal axial load at d (N)	F_a	1729.5890	
Nominal radial load at d (N)	F_r	1489.5680	
Nominal normal load at d (N)	F_n	4355.2070	
Nominal tangential load per unit facewidth (N/mm)	w	105.9750	
Circumferential speed at d (m/s)	v	40.4410	
Sliding velocity at Tip dia. (m/s)	v_g	6.9060	7.3250
Nominal tangential load at d_w (N)	F_{tw}	3709.1150	
Nominal axial load at d_w (N)	F_{aw}	1729.5890	
Nominal radial load at d_w (N)	F_{rw}	1489.5680	
Circumferential speed at d_w (m/s)	v_w	40.4410	
Running-in value (um)	y_p	0.5930	
Running-in value (um)	y_f	0.6370	
Effective single pitch & profile deviation (um)	f_{pb_eff}	7.3080	
Effective single pitch & profile deviation (um)	f_{fa_eff}	7.8620	
Correction factor	C_M	0.8000	
Gear blank factor	C_R	0.8980	
Basic rack factor	C_B	0.9750	
Material factor	E/E_{st}	1.0000	
Theoretical single stiffness (N/mm/um)	c'_{th}	18.7930	
Single tooth stiffness (N/mm/um)	c'	11.9300	
Mesh stiffness (for K_v , $KH\alpha$, $KF\alpha$), (N/mm/um)	$c_{y\alpha}$	15.9230	
Mesh stiffness (for $KH\beta$, $KF\beta$), (N/mm/um)	$c_{y\beta}$	13.5350	
Mean diameter for K_v (mm)	$d_m (K_v)$	76.7360	167.2130
Moment of inertia for K_v (kg.mm ²)	$J (K_v)$	732.5830	9872.9980
Moment of gear inertia per unit facewidth (kg.mm)	J^*	20.9310	282.0860
Relative gear mass per unit facewidth (kg/mm)	m^*	0.0163	0.0466
Relative gear mass per unit facewidth (kg/mm)	m_{red}	0.0121	
Resonance running speed (rpm)	n_{E1}	9908.1570	
Resonance ratio	N	1.0090	
Resonance ratio in main resonance ratio	N_s	0.8500	
Resonance range	-	main resonance range	
Dynamic factor	K_v	1.4870	

	Symbol	Pair1 Gear1	Pair1 Gear2
Face load factor - flank	$K_{H\beta}$	1.3000	
Face load factor - root	$K_{F\beta}$	1.2560	
Transverse load factor - flank	$K_{H\alpha}$	1.1390	
Transverse load factor - root	$K_{F\alpha}$	1.1390	
Number of load cycles	N_L	6.0000×10^9	2.7630×10^9

3.2. Pitting

S-N curve (Pitting), Load case 1, Meshing pair: Gear1 - Gear2



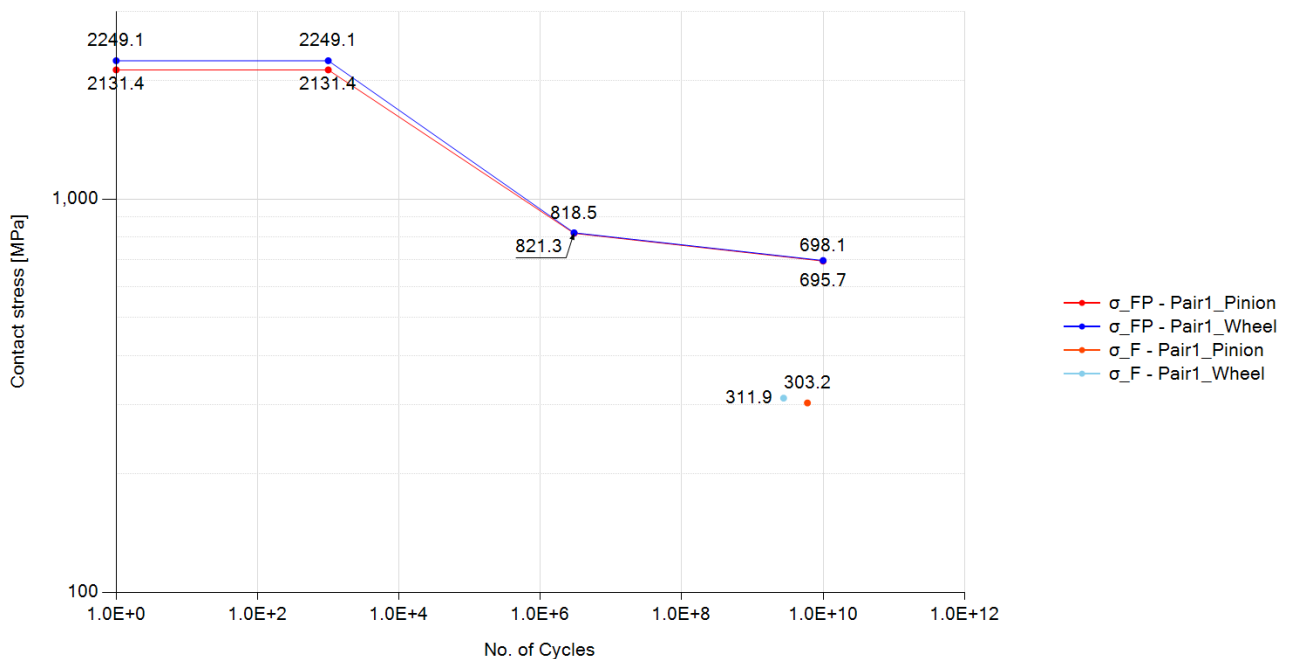
Load case 1, Meshing pair: Gear1 - Gear2

	Symbol	Pair1 Gear1	Pair1 Gear2
Rating Method	-	ISO 6336-2:2019	
Contact ratio factor	Z_ϵ	0.8320	
Zone factor	Z_H	2.3040	
Elasticity factor	Z_E	189.8120	
Young's modulus for tooth flank (N/mm ²)	E	2.0600×10^5	2.0600×10^5
Helix angle factor	Z_β	1.0500	
Single pair tooth contact factor (pinion)	Z_B	1.0000	
Single pair tooth contact factor (wheel)	Z_D	1.0000	
Nominal contact stress at the pitch point (N/mm ²)	σ_{H0}	540.6950	
Contact stress (N/mm ²)	σ_H	896.8840	896.8840
Life factor	Z_{NT}	0.8630	0.8840

	Symbol	Pair1 Gear1	Pair1 Gear2
Lubricant factor	Z_L	0.9460	0.9460
Velocity factor	Z_V	1.0410	1.0410
Roughness factor	Z_R	0.9630	0.9630
Work hardening factor	Z_W	1.0000	1.0000
Size factor	Z_X	1.0000	1.0000
Permissible contact stress - static (N/mm ²)	σ_{HP_static}	2400.0000	2400.0000
Permissible contact stress - mid point (N/mm ²)	σ_{HP_mid}	0.0000	0.0000
Permissible contact stress - reference (N/mm ²)	σ_{HP_ref}	1421.8020	1421.8020
Permissible contact stress - long (N/mm ²)	σ_{HP_long}	1208.5320	1208.5320
Permissible contact stress (N/mm ²)	σ_{HP}	1227.6170	1257.1650
Safety factor for Contact	S_H	1.3690	1.4020
Contact damage (%)	-	0.0040	0.0020
Cycle to failure	-	1.6690×10^{14}	1.6690×10^{14}

3.3. Bending

S-N curve (Bending), Load case 1, Meshing pair: Gear1 - Gear2



Load case 1, Meshing pair: Gear1 - Gear2

	Symbol	Pair1 Gear1	Pair1 Gear2
Rating Method	-	ISO 6336-3:2019	
Tooth form factor	Y_F	1.0220	0.9890
Stress correction factor	Y_S	1.9800	2.1050

	Symbol	Pair1 Gear1	Pair1 Gear2
Working angle (deg)	α_{Fen}	19.0430	19.7570
Bending moment arm (mm)	h_{Fe}	2.0170	2.1650
Tooth thickness at root (mm)	s_{Fn}	4.2640	4.4810
Tooth root radius (mm)	ρ_F	1.0830	0.9720
Helix angle factor	Y_{β}	1.0630	
Deep tooth factor	Y_{DT}	1.0000	
Rim thickness factor	Y_B	1.0000	1.0000
Nominal tooth root stress (N/mm ²)	σ_{F0}	114.0290	117.2930
Tooth root stress (N/mm ²)	σ_F	303.1860	311.8660
Life factor	Y_{NT}	0.8590	0.8720
Relative notch sensitive factor	$Y_{\delta relT}$	0.9950	0.9980
Surface factor	Y_{RrelT}	0.9570	0.9570
Size factor	Y_X	1.0000	1.0000
Stress correction factor	Y_{ST}	2.0000	
Mean stress influence factor	Y_M	1.0000	1.0000
Technical factor	Y_T	1.0000	1.0000
Permissible tooth root stress - static (N/mm ²)	σ_{FP_static}	2131.3960	2249.1360
Permissible tooth root stress - reference (N/mm ²)	σ_{FP_ref}	818.4680	821.2500
Permissible tooth root stress - long (N/mm ²)	σ_{FP_long}	695.6970	698.0630
Permissible tooth root stress (N/mm ²)	σ_{FP}	702.8540	716.2850
Safety factor for Bending	S_F	2.3180	2.2970
Bending damage (%)	-	0.0000	0.0000
Cycle to failure	-	5.1340×10^{20}	1.4860×10^{20}

3.4. Scuffing

Load case 1, Meshing pair: Gear1 - Gear2

	Symbol	Pair1 Gear1	Pair1 Gear2
Lubrication type	-	Spray lubrication	
Lubricant temperature (deg)	θ_{oil}	80.0000	
Lubricant viscosity (mPa.s)	η_{oil}	13.1510	
Load stage according to FZG A/8, 3/90	S_{FZG}	12.0000	
Gear density (kg/m ³)	ρ_M	7830.0000	7830.0000
Gear specific heat (J/kg.K)	c_M	485.0000	485.0000
Gear heat conductivity (N/s.K)	λ_M	50.0000	50.0000
Gear roughness (μm)	R_a	0.6000	0.6000
Number of mating gears	n_p	1.0000	
Effective facewidth (mm)	b_{eff}	29.7500	

	Symbol	Pair1 Gear1	Pair1 Gear2
Relevant tip relief (μm) (Min. value = 2 μm)	C_a	2.0000	2.0000
Transverse unit load (N/mm)	w_Bt	291.5890	
Lubrication system factor	X_S	1.2000	
Multiple meshing factor	X_mp	1.0000	
Structural factor	X_W	1.0000	1.0000
Structural factor (total)	X_Wtot	1.0000	
Thermal contact factor	B_M	13.7800	13.7800
Multiple path factor	K_mp	1.0000	
Optimal tip relief (μm)	C_eff	8.3190	
Angle factor	X_αβ	0.9630	
Lubricant factor	X_L	0.8790	
Roughness factor	X_R	0.7750	
Mean coefficient of friction	μ_m	0.0410	
Maximum flash temperature (deg)	θ_fl_max	51.0670	
Mean flash temperature (deg)	θ_fl_mean	19.3600	
Total mass temperature (deg)	θ_M	90.9190	
Scuffing temperature (deg)	θ_S	364.8370	
Maximum contact temperature (deg)	θ_Bmax	141.9860	
Safety factor for flash temperature	S_B	4.5950	

3.5. Micro-pitting

Load case 1

	Symbol	Gear1 - Gear2
minimum lubricant film thickness (μm)	h_Ymin	0.2806
minimum specific lubricant film thickness in the contact area	λ_GFmin	0.4676
permissible specific lubricant film thickness	λ_GFP	0.2111
safety factor against micropitting	S_λ	2.2154
transverse tangential load at reference cylinder per mesh (N)	F_t	3709.1150
transverse load in plane of action (N)	F_bt	3997.0430
reduced modulus of elasticity (N/mm ²)	E_r	2.2637×10 ⁵
thermal contact coefficient of pinion (N/(ms ^{0.5} *K))	B_M1	13779.6040
thermal contact coefficient of wheel (N/(ms ^{0.5} *K))	B_M2	13779.6040
failure load stage according to FVA 54/7 (90°)	-	0.0000
effective arithmetic mean roughness value (μm)	R_a	0.6000
roughness factor	X_R	1.0700
lubricant factor	X_L	1.0000
lubrication factor	X_S	1.2000

	Symbol	Gear1 - Gear2
elasticity factor (N/mm ²) ^{0.5}	Z_E	189.8120
helical load factor	K_B γ	1.3000
load losses factor	H_v	0.0860
mean coefficient of friction	μ_m	0.0440
profile position at min. lubricant film thickness	-	0.0000
tip relief factor	X_Ca	1.0000
load sharing factor	X_A	0.8990
buttressing factor	X_but,A	1.3000
effective tip relief (μm)	C_eff	0.0000
Applied tip relief in calculation (μm)	C_a	0.0000
sum of tangential velocities at point A (m/s)	$v_{\Sigma,A}$	27.4360
sliding velocity at point A (m/s)	$v_{g,A}$	-7.3250
tangential velocity on pinion at point A (m/s)	$v_{r1,A}$	10.0560
tangential velocity on wheel at point A (m/s)	$v_{r2,A}$	17.3810
oil inlet/sump temperature (°C)	θ_{oil}	80.0000
bulk temperature (°C)	θ_M	92.0060
flash temperature at point A (°C)	$\theta_{fl,A}$	56.9930
contact temperature at point A (°C)	$\theta_{B,A}$	148.9990
pressure-viscosity coefficient at 38 °C (m ² /N)	α_{38}	1.8390×10^{-8}
pressure-viscosity coefficient at bulk temperature (m ² /N)	$\alpha_{\theta M}$	1.3880×10^{-8}
pressure-viscosity coefficient at point A contact temperature (m ² /N)	$\alpha_{\theta B,A}$	1.0370×10^{-8}
kinematic viscosity at 40 °C (mm ² /s)	ν_{40}	68.0000
kinematic viscosity at 100 °C (mm ² /s)	ν_{100}	8.8000
kinematic viscosity at point A contact temperature (mm ² /s)	$\nu_{\theta B,A}$	3.5390
dynamic viscosity at 38 °C (Ns/m ²)	η_{38}	0.0650
dynamic viscosity at bulk temperature (Ns/m ²)	$\eta_{\theta M}$	0.0090
dynamic viscosity at oil inlet/sump temperature (Ns/m ²)	$\eta_{\theta oil}$	0.0120
dynamic viscosity at point A contact temperature (Ns/m ²)	$\eta_{\theta B,A}$	0.0030
normal radius of relative curvature at point A (mm)	$\rho_{n,A}$	8.2620
material parameter	G_M	3141.2990
local velocity parameter at point A	U_A	6.5230×10^{-11}
local load parameter at point A	W_A	0.0002
local sliding parameter at point A	S_GF,A	0.2350
local nominal Hertzian contact stress at point A (Method B) (N/mm ²)	$p_{H,A}$	669.0560
local Hertzian contact stress at point A including the load factors K (Method B) (N/mm ²)	$p_{dyn,A}$	1109.8060

4. Power loss

4.1. Mesh loss

Load case 1

	Symbol	Gear1 - Gear2
Rating Method	-	ISO/TR 14179-2:2001
Input Power (kW)	P	150.0000
Gear ratio	u	2.1714
Tangential speed at working pitch dia. (m/s)	v_{wt}	40.4409
Tangential force at pitch dia. (N)	F_t	3709.1151
Tooth normal force, transverse section (N)	F_{bt}	3997.0425
Arithmetic average roughness of pinion	Ra1	0.6000
Arithmetic average roughness of wheel	Ra2	0.6000
Roughness factor	X_R	0.6000
Sum velocity (m/s)	v_{Σ}	30.1420
Equivalent radius of curvature (mm)	ρ	10.7368
Kinematic viscosity of oil at operating temp. (cSt)	η_{oil}	14.8602
Lubricant type	-	Mineral oil base
Lubricant factor	X_L	1.0000
distributed force on tooth surface(N/m)	F/b	150.0000
Mean coefficient of friction of the gear mesh	μ_{mz}	0.0317
Tooth loss factor	H_v	0.0858
Mesh power loss (W)	P_{VZP}	407.2432
Mesh efficiency (%)	η_z	99.7285

4.2. Churning loss

Load case 1

	Symbol	Gear1	Gear2
Rating Method	-	Changenet (2011)	Changenet (2011)
Rotating speed (rpm)	n	10000.0000	4605.2632
Lubricant	-	Oil: ISO-VG 68	Oil: ISO-VG 68
Gear immersed depth (mm)	h_e	38.6182	83.8567
Oil density (kg/m ³)	ρ	885.0000	885.0000
Operating Viscosity (cSt)	ν	14.8602	14.8602
dynamic viscosity (kg/m.s)	μ	0.0000	0.0000
Reynolds number	Re	95249.8840	95249.8840
Froude number	Fr	4316.9852	1988.0853
Centrifugal acceleration (m/s ²)	γ	15276.3885	4195.4458

	Symbol	Gear1	Gear2
Tip diameter of gear (mm)	d_a	81.2365	171.7134
Pitch diameter of gear (mm)	d_p	77.2365	167.7134
Facewidth (mm)	b	35.0000	35.0000
Dimensionless drag torque	C_m	0.0009	0.0020
Immersion area of the gear (m ²)	S_m	0.0154	0.0454
Total oil churning loss (W)	P	387.0727	2633.7660
Total oil churning torque (N.m)	T	0.3696	5.4613

4.3. Windage loss

Load case 1

	Symbol	Gear1	Gear2
Rating Method	-	None	None
Windage loss (W)	P_WL	0.0000	0.0000
Windage torque (N.m)	T_WL	0.0000	0.0000